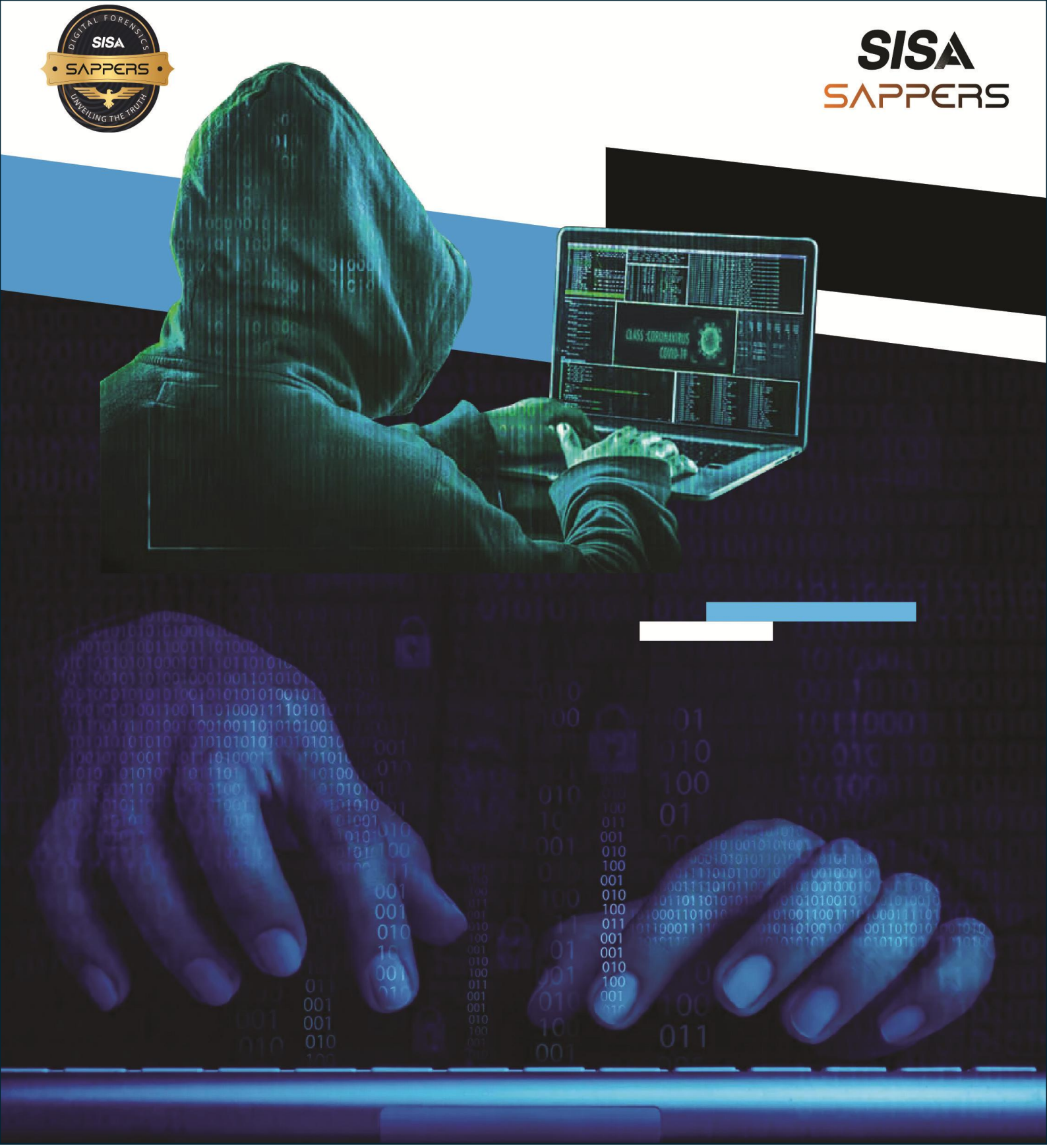

Payment Intelligence Report – Fraud & Scam Landscape (The Shift to Agentic Commerce, Upstream Identity Fraud, and Post-Quantum)

Executive Summary

The opening weeks of 2026 signal a fundamental paradigm shift in the payment security landscape. We are witnessing the institutionalization of **"Agentic Commerce,"** moving the primary merchant challenge from blocking bots to authenticating authorized AI agents. Concurrently, fraud vectors are migrating **"upstream,"** transitioning from transaction-level theft to total identity-level compromise via deepfake-enabled scams.

This report details the emergence of physical infrastructure threats like **SMS Blasters**, the regulatory shift toward **dynamic authentication** in major markets like India, and the critical multi-year migration to **Post-Quantum Cryptography (PQC)**. To maintain parity with high-velocity threats, financial institutions must move beyond static defenses toward real-time, cryptographic, and behavioral trust frameworks.

1. Agentic Commerce & The "Cryptographic Handshake"

1.1 The Identity Crisis: Legitimate Agents vs Malicious Bots

By early 2026, AI software agents—capable of executing autonomous purchases—have entered the mainstream. However, these agents often trigger legacy fraud filters by mimicking the high-velocity behaviors typically associated with credential stuffing or Account Takeover (ATO).

Technical Pattern: The "Trusted Agent Protocol" (TAP)

To bridge the trust gap, industry leaders are adopting standardized cryptographic verification frameworks:

Protocol	Developer / Standard	Security Mechanism
Trusted Agent Protocol (TAP)	Visa / Cloudflare	RFC 9421 (HTTP Message Signatures) for time-bound requests.
Agent Pay Acceptance	Mastercard / Google	"Agentic Tokens"—dynamic credentials with embedded trust signals.
Agentic Commerce Protocol (ACP)	Stripe / OpenAI	Open-source specifications for programmatic commerce flows.

- **Key Risks:** Malicious actors are already attempting to "spoof" agent headers to bypass CDN-layer bot mitigation. Furthermore, the legal liability for "unintended" agent purchases remains a significant gray area between platforms and merchants.

1.2 Detection Strategies

- **Behavioral Recalibration:** Fraud engines are being retrained to recognize "healthy" bot patterns, such as multi-merchant cart orchestration, while flagging price-scraping or DDoS-level volume.
- **Cryptographic Validation:** Implementation of Signature-Input and Signature-Agent headers to validate agent intent before a checkout flow is initiated.

2. The AI-Driven Identity Battle: Upstream Fraud

2.1 The "Phantom Hacker" 3-Step Chain

Generative AI has industrialized "Phantom Hacker" scams, moving fraud from the point of sale to the point of identity.

1. **The Lure:** AI-authored alerts notify users of a fabricated security breach.
2. **The Hook:** Hyper-realistic deepfake voice or video calls convince victims to move funds to "protected" accounts.
3. **The Drain:** Once "upstream" access to the banking identity is secured, attackers execute Authorized Push Payments (APP) or Account-to-Account (A2A) transfers.

2.2 Identity vs. Transactional Compromise

Fraud is no longer about stealing a card number; it is about **owning the profile.** When an attacker compromises on an identity via synthetic IDs or deep impersonation, transaction-level rules (e.g., spending limits) become ineffective because the attacker possesses the "authority" to override them.

3. Infrastructure & Network Threats

3.1 SMS Blasters: Bypassing the Carrier

Criminal organizations are increasingly deploying **"SMS Blasters"** (localized IMSI catchers). These devices mimic legitimate cell towers to force nearby mobile devices to connect.

- **Impact:** Messages bypass carrier-level security filters, delivering smishing lures directly to devices with high-trust signals.
- **Indicator:** Users may experience a brief "No Service" state followed by immediate receipt of high-fidelity, urgent retail or tax alerts.

3.2 Post-Quantum Cryptography (PQC) Migration

Following the finalization of NIST's FIPS standards (203, 204, and 205), the migration to quantum-resistant algorithms has begun.

- **The "Harvest Now, Decrypt Later" Threat:** Attackers are harvesting encrypted payment data today, intending to decrypt it once quantum computing matures.
- **Operational Impact:** PQC algorithms like **ML-KEM** require larger keys and signatures, increasing secure connection bandwidth by **15-20%**.

4. Regulatory Cybersecurity & Compliance

4.1 India's RBI Authentication Reset (April 2026)

The Reserve Bank of India (RBI) has mandated a shift away from SMS-based OTPs, citing vulnerabilities to SIM-swapping.

- **Dynamic Factors:** For non-card-present transactions, at least one authentication factor must be created dynamically for that specific transaction.
- **Contextual Intelligence:** Issuers must now evaluate IP geolocation, device reputation, and historical behavior in real-time.

4.2 Cross-Border Real-Time Compliance (UPI-TIPS)

The linkage between India's **UPI** and Europe's **TIPS** system is entering the realization phase.

- **The Challenge:** Screening systems must perform sanctions checks and fraud detection within seconds across the Euro-India corridor, requiring automated AML/CFT rule application at the point of origin.

5. Strategic Recommendations

- **For Infrastructure Providers:** Inventory all RSA/ECC dependencies and begin firmware updates to support **ML-KEM** and **ML-DSA** for long-term data protection.
- **For Merchants:** Adopt **RFC 9421** HTTP signatures to distinguish between "trusted" AI agents and malicious bots.
- **For Compliance Officers:** Implement **"Know Your Agent" (KWA)** procedures. Update due diligence to focus on the authorization chain between the human user and their software agent.

Conclusion

As we navigate 2026, cybersecurity is no longer a peripheral technical concern, it is the foundational pillar of global commerce. The convergence of **Agentic Commerce**, **SMS Blaster technology**, and **Quantum-readiness** demands a "resilience-first" mindset. Success in this new era requires moving beyond static prevention toward a dynamic, cryptographic trust model that can authenticate intent and identity in real-time, across every layer of the payment ecosystem.